Equations

Week 3A

Overview

Linear and Quadratic Equations

Key Ideas - Solving Linear Equations

- Solving Equations with Fractions

- Introduction to Non-Linear Equations

- Introduction to Quadratic Equations

- Solving Quadratic Equations by Factorisation

Key Idea – Solving Linear Equations

Many Engineering problems can be solved by using an equation or a set of equations.

An **equation is a mathematical statement containing one or more unknown variables and an equal sign**, indicating that the two quantities involved are equal.

An equation can be solved to find the value of the unknown, for example in the equation

$$2x - 3 = 9$$

the expression on the left hand side ($2x - 3$) is equal to the number on the right hand side (9), this equation can be solved to find the value of x for which this statement is true.

What is a Linear Equation?

A linear equations is an equation that can be written in the form

$$\boldsymbol{ax + b = c} \quad a \neq 0$$

where *a, b* and *c* have known values and *x* is the unknown variable to be evaluated.

Examples:

$$2x + 3 = 7$$
$$-4x + 5 = 0$$
$$-\frac{x}{3} + 3 = 2$$
$$-x + 5 = 0$$

The important thing to recognise in a linear equation is that the **power of the unknown, *x*, is 1.**

Solving Linear Equations

Discussion Questions: Consider the linear equation $2x - 3 = 9$

Check that $x = 6$ is a solution.

$$\begin{aligned} 2x - 3 &= 2 \times 6 - 3 \\ &= 12 - 3 \\ &= 9 \end{aligned}$$

$x = 6$ is a solution

Check that $x = 4$ is not a solution.

$$\begin{aligned} 2x - 3 &= 2 \times 4 - 3 \\ &= 8 - 3 \\ &= 5 \\ &\neq 9 \end{aligned}$$

$x = 4$ is not a solution

To **solve an equation means to find all values of the variable for which the statement is true**. This value is called the **solution** of the equation and the value is said to **satisfy** the equation. To check if the solution is correct, **substitute** the value for the variable into the equation and see if **both sides are equal**.

To **solve an equation** we need to **get the variable on its own**, which means that we make it the **subject of the equation**. To make the variable the subject of the equation, we **need to "undo" whatever has been done to it**. We undo each operation by applying the **inverse operation** (the opposite).

Discussion Questions: How do we undo each of the following operations?

Operation	Inverse Operation
$+$	$-$
$-$	$+$
$\times$	$\div$
$\div$	$\times$
$(\ )^2$	$\sqrt{\ }$
$(\ )^3$	$\sqrt[3]{\ }$
$\sqrt{\ }$	$(\ )^2$
$\sqrt[3]{\ }$	$(\ )^3$

When we apply an inverse operation we need to **maintain the equality** of the equation (that is, keep the equation **balanced**) by applying the inverse operation to both sides of the equation.

Specifically, the following operations can be used in the process of solving an equation:

- Completely **swap** left hand expression with right hand expression.
- **Add/subtract** a number or expression to/from **both sides**.
- **Multiply/divide both sides** by a number or an expression.

Note: *Whatever you do to one side of an equation you must do to the other side.*

Examples:

Solve the following equations:

$$2x+5=11$$

$2x=6$ Subtract 5 from both sides

$x=3$ Divide both sides by 2

$$5-4x=3$$

$-4x=-2$ Subtract 5 from both sides

$x=\frac{1}{2}$ Divide both sides by -4

$-\frac{x}{4}=2$ Multiply both sides by -4

$$x=-8$$

$$\frac{3x}{5}=6$$

$3x=30$ Multiply 5 to both sides

$x=10$ Divide both sides by 3

$\frac{x}{3}+5=-2$ Subtract 5 from both sides

$\frac{x}{3}=-7$ Multiply both sides by 3

$$x=-27$$

$\frac{2}{3}(x+1)=-5$ Multiply both sides by 3

$2x+2=-15$ Subtract 2 from both sides

$2x=-17$ Divide both sides by 2

$$x=-\frac{17}{2}$$

Exercise: Solve the following equations:

(a) $1 = 4x - 2$ $\quad (+2)$

$3 = 4x$ $\quad$ (swap)

$4x = 3$ $\quad (\div 4)$

$x = \frac{3}{4}$

(b) $\dfrac{3(x+1)}{4} - 7 = 5$ $\quad (+7)$

$\dfrac{3(x+1)}{4} = 12$ $\quad (\times 4)$

$3(x+1) = 48$ $\quad (\div 3)$

$x+1 = 16$ $\quad (-1)$

$x = 15$

Discussion Questions:

What if the variable appears in more than one place in the equation, such as in $2n + 4 = 5n - 3$?

subtract 2n from both sides

$4 = 3n - 3$

add 3 to both sides

$7 = 3n$

swap $3n = 7$

divide both sides by 3

$n = \frac{7}{3}$

Summary, when solving an equation:

- If the variable appears in more than one place in the equation, manipulate the equation so it appears once.
- Use inverse processes to isolate the variable on one side of the equation.

Examples:

Solve the following equations:

$9x - 7 = 4x + 28$	
$9x = 4x + 35$	Add 7 to both sides
$5x = 35$	Subtract $4x$ from both sides
$x = 7$	Divide both sides by 5

$3(2a + 4) = 4a + 16$	
$6a + 12 = 4a + 16$	Expand the bracket
$2a + 12 = 16$	Subtract $4a$ from both sides
$2a = 4$	Subtract 12 from both sides
$a = 2$	Divide both sides by 2

Checking solution for $3(2a + 4) = 4a + 16$:

Substitute $a = 2$ and check that LHS = RHS.

$LHS = 3(2 \times 2 + 4) = 24$

$RHS = 4 \times 2 + 16 = 24$

LHS = RHS $\quad \therefore$ a = 2 is a valid solution

Exercise: Solve the following equations:

(a) $5 - x = 27 + x$ $(+x)$

$5 = 27 + 2x$ (-27)

$-22 = 2x$ (swap)

$2x = -22$ $(\div 2)$

$x = -11$

(b) $40 - 5(7 - x) = 3(x + 1)$ (expand)

$40 - 35 + 5x = 3x + 3$

$5 + 5x = 3x + 3$ $(-3x)$

$5 + 2x = 3$ (-5)

$2x = -2$ $(\div 2)$

$x = -1$

Key Idea – Solving Linear Equations with Fractions

So far, we have solved equations with only one fraction involved. We will now look at examples that involve more than one fraction.

Discussion Questions:

How do we solve the equation $\frac{x}{2}+\frac{x}{5}=1$ that has more than 1 fraction?

Make a common denominator

$$\frac{5x}{5\times 2}+\frac{2x}{2\times 5}=1 \rightarrow \frac{5x+2x}{10}=1 \rightarrow \frac{7x}{10}=1 \rightarrow 7x=10 \rightarrow x=\frac{10}{7}$$

Would this method work as easily for the equation $\frac{x}{2}+3=\frac{x}{5}$?

No

There are a several different approaches that can be used to solve an equation with more than one fraction, as shown in the following examples.

Example : Solve this equation, by finding the lowest common denominator: $\frac{x+2}{3}=\frac{5x}{2}+4$

$$\frac{x+2}{3}=\frac{5x}{2}+4$$

$$\frac{2(x+2)}{2\times3}=\frac{3\times5x}{3\times2}+\frac{6\times4}{6\times1}$$ Make a common denominator of 6

$$\frac{2(x+2)}{6}=\frac{15x}{6}+\frac{24}{6}$$

$$\frac{2(x+2)}{6}=\frac{15x+24}{6}$$

$$2(x+2)=15x+24$$ Multiply both sides by 6

$$2x+4=15x+24$$ Expand the brackets

$$-13x+4=24$$ Subtract $15x$ from both sides

$$-13x=20$$ Subtract 4 from both sides

$$x=-\tfrac{20}{13}$$

$$=-1\tfrac{7}{13}$$ Divide both sides by -13

You may prefer to **multiply each term by the lowest common denominator (LCD) first, to eliminate the fractions.** We then continue to use inverse processes to make the variable the subject.

***Example*:** Solve the following equation:

$$\frac{x+2}{3} = \frac{5x}{2} + 4$$ Find the lowest common denominator (LCD). LCD = 6

$$6\left(\frac{x+2}{3}\right) = 6\left(\frac{5x}{2} + 4\right)$$ Multiply each term by 6 (LCD) to eliminate the fractions

$$6\left(\frac{x+2}{3}\right) = 6\left(\frac{5x}{2}\right) + 6 \times 4$$

$$2(x+2) = 3(5x) + 24$$

$$2x + 4 = 15x + 24$$ Expand the brackets

$$-13x + 4 = 24$$ Subtract $15x$ from both sides

$$-13x = 20$$ Subtract 4 from both sides

$$x = -\tfrac{20}{13}$$

$$= -1\tfrac{7}{13}$$ Divide both sides by -13

Exercise: Solve the following equations:

(a) $\dfrac{x+1}{2}-\dfrac{x}{3}=\dfrac{7x}{4}$

$$12\times\left(\frac{x+1}{2}-\frac{x}{3}\right)=12\times\left(\frac{7x}{4}\right)$$

$$\frac{12(x+1)}{2}-\frac{12x}{3}=\frac{12\times 7x}{4}$$

$$6(x+1)-4x=3\times 7x$$

$$6x+6-4x=21x$$

$$2x+6=21x \qquad (-2x)$$

$$6=19x \qquad (\text{swap})$$

$$19x=6 \qquad (\div 19)$$

$$x=\frac{6}{19}$$

(b) $\dfrac{x+3}{4}-\dfrac{2x-1}{6}=x+1$

$$12\times\left(\frac{x+3}{4}-\frac{2x-1}{6}\right)=12\times(x+1)$$

$$\frac{12(x+3)}{4}-\frac{12(2x-1)}{6}=12x+12$$

$$3(x+3)-2(2x-1)=12x+12$$

$$3x+9-4x+2=12x+12$$

$$-x+11=12x+12 \quad (+x)$$

$$11=13x+12 \quad (-12)$$

$$-1=13x \quad (\text{swap})$$

$$13x=-1 \quad (\div 13)$$

$$x=-\frac{1}{13}$$

In equations where there is **just a fraction** on each side of the equals sign, there is a shortcut that can be taken.

Discussion Questions:

Solve the equation $\frac{3x}{2}=\frac{x-1}{7}$ using the technique we have just learnt.

$$14\times\frac{3x}{2}=14\times\frac{(x-1)}{7}$$

$$7\times 3x = 2(x-1)$$

$$21x = 2x-2$$

$$19x = -2$$

$$x = -\frac{2}{19}$$

Can you think of a short cut to solve $\frac{3x}{2}=\frac{x-1}{7}$?

$$7(3x) = 2(x-1)$$

In equations where there is just a fraction on each side of the equals sign, this short cut approach is referred to as **"cross multiplication".**

Examples**:** Solve the equation

$$\frac{8x}{7} = \frac{2x-1}{5}$$

$5(8x) = 7(2x-1)$ Cross-multiply [equivalent to multiplying both sides by 35]

$40x = 14x - 7$ Expand the brackets

$26x = -7$ Subtract $14x$ from both sides

$x = -\frac{7}{26}$ Divide both sides by 26

Note:

Cross multiplication allows us to leave out this first step of working $35\left(\frac{8x}{7}\right) = 35\left(\frac{2x-1}{5}\right)$

Exercise:

Solve the following equation: $\dfrac{4-3x}{4}=\dfrac{x+6}{3}$

$$3(4-3x)=4(x+6)$$

$$12-9x=4x+24 \qquad (+9x)$$

$$12=13x+24 \qquad (-24)$$

$$-12=13x \qquad (\text{swap})$$

$$13x=-12 \qquad (\div 13)$$

$$x=\frac{-12}{13}$$

What if the Variable is in the Denominator?

To solve an equation with the variable in the denominator we apply **inverse operations** in the same way as we do when solving a linear equation, remembering to keep the **equation balanced**.

Note, the solutions will have restrictions because some values would make the denominator zero and division by zero is undefined.

Examples**:** Solve the equation $\dfrac{3}{x-4}=\dfrac{2}{x-2}$

$$\frac{3}{x-4}=\frac{2}{x-2},\ x\neq 4,\ x\neq 2$$

$\dfrac{3(x-4)(x-2)}{(x-4)}=\dfrac{2(x-4)(x-2)}{(x-2)}$ Cross-multiply [equivalent to multiplying both sides by $(x-4)(x-2)$]

$3(x-2)=2(x-4)$ Expand the brackets

$3x-6=2x-8$ Subtract $2x$ and add 8 to/from both sides

$x=-2$

Examples**:** Solve this equation by

Solve this equation by <u>finding the lowest common denominator</u>: $\frac{3}{x}+2=\frac{1}{x}$

$$\frac{3}{x}+2=\frac{1}{x},\quad x\neq 0$$

$$x\left(\frac{3}{x}\right)+2\times x=x\left(\frac{1}{x}\right)$$ LCD = x. Multiply each term by x to eliminate the fractions

$$3+2x=1$$

$$2x=-2$$

$$x=-1$$

Solve the equations by <u>rearranging and simplifying</u> the fractions: $\frac{3}{x}+2=\frac{1}{x}$

$$\frac{3}{x}+2=\frac{1}{x},\quad x\neq 0$$

$$\frac{3}{x}-\frac{1}{x}=-2$$ Subtract $\frac{1}{x}$ and 2 from both sides

$$\frac{2}{x}=-2$$

$$2=-2x$$

$$x=-1$$

Exercise:

Solve the following equations:

(a) $\dfrac{3x-2}{2x-1}=\dfrac{3x+1}{2x+3}$

$$(2x+3)(3x-2) = (3x+1)(2x-1)$$

$$6x^2-4x+9x-6 = 6x^2-3x+2x-1$$

$$6x^2+5x-6 = 6x^2-x-1 \qquad (-6x^2)$$

$$5x-6 = -x-1 \qquad (+x)$$

$$6x-6 = -1 \qquad (+6)$$

$$6x = 5 \qquad (\div 6)$$

$$x = \frac{5}{6}$$

(b) $\frac{1}{x}-\frac{1}{2x}+\frac{1}{3x}=5$

$6x\left(\frac{1}{x}-\frac{1}{2x}+\frac{1}{3x}\right)=6x\times 5$

$\frac{6\cancel{x}}{\cancel{x}}-\frac{6\cancel{x}}{2\cancel{x}}+\frac{6\cancel{x}}{3\cancel{x}}=30x$

$6-3+2=30x$

$5=30x$ (swap)

$30x=5$ (÷30)

$x=\frac{5}{30}$

$x=\frac{1}{6}$

Key Idea – Introduction to Non-Linear Equations

A non-linear equation is different from a linear equation as the **power of the unknown, *x*, is NOT 1**. So, we are turning our attention to equations involving indices. The first non-linear equation we will consider is a **Quadratic Equation.**

Key Idea – Introduction to Quadratic Equations

A **quadratic equation** is an equation which can be written in the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

where the values of a, b and c are given and x is the variable.

To solve a quadratic equation, we must **find values of the variable *x*** which make the **left-hand side equal to the right-hand side**. Such values are known as the **roots** or **solutions** of the quadratic equation. A quadratic equation will generally have **two solutions** but may have **one solution** or **no solutions**.

There are **three methods** for solving quadratic equations:

- **Factorising**
- **Completing the Square**
- **Using the Quadratic Formula**

Key Idea – Solving Quadratic Equations by Factorisation

The easiest way to solve a quadratic equation is to **factorise the quadratic expression**. However, we need to be aware **that not all quadratic expressions can be factorised.**

***Example*: Solve $2x^2 = 72$**

$$
\begin{aligned}
2x^2 &= 72 \\
2x^2 - 72 &= 0 \\
2(x^2 - 36) &= 0 \\
2(x-6)(x+6) &= 0 \\
(x-6)(x+6) &= 0 \\
(x-6) = 0 \quad &\text{or} \quad (x+6) = 0 \\
x &= 6, \ -6
\end{aligned}
$$

OR

$$
\begin{aligned}
2x^2 &= 72 \\
x^2 &= \frac{72}{2} \\
x^2 &= 36 \\
x &= \pm\sqrt{36} \\
x &= \pm 6
\end{aligned}
$$

Example: Solve $3x^2 + 7x - 6 = 0$

$3x^2 + 7x - 6 = 0$ — Put all terms on the left hand side.

$(3x-2)(x+3) = 0$ — Factorise the left hand side.

$3x - 2 = 0 \quad \text{or} \quad x + 3 = 0$ — Here either $(3x\text{-}2) = 0$ or $(x+3) = 0$ for the equation to be true.

$x = \frac{2}{3} \quad \text{or} \quad x = -3$ — Solve for x in both cases.

Example: Solve the following equation by first expanding both sides.

$$(x-3)(2x+1) = x(x-5)$$

$$2x^2 + x - 6x - 3 = x^2 - 5x$$

$$2x^2 - 5x - 3 = x^2 - 5x$$

Expand both sides.

$$2x^2 - x^2 - 5x + 5x - 3 = 0$$

Put all terms on the left hand side.

$$x^2 - 3 = 0$$

Collect like terms.

$$x^2 = 3$$

Make x the subject.

$$x = \pm\sqrt{3}$$

Take the square root of both sides.

Exercise:

Use factorisation to solve each of the following quadratic equations:

(a) $5x^2 - 125 = 0$ $\rightarrow$ $5(x^2-25)=0$

$5(x-5)(x+5)=0$ $\rightarrow$ $x=5$ or $x=-5$

$x-5=0$ $\quad$ $x+5=0$

$x=5$ $\quad$ $x=-5$

(b) $x^2 - 4x = 0$ $\rightarrow$ $x(x-4)=0$ $\rightarrow$ $x=0$ or $x=4$

$x=0$ $\quad$ $x-4=0$

$x=4$

(c) $x^2 - 6x - 7 = 0$ $\rightarrow$ $(x-7)(x+1)=0$ $\rightarrow$ $x=7$ or $x=-1$

$x-7=0$ $\quad$ $x+1=0$

$x=7$ $\quad$ $x=-1$